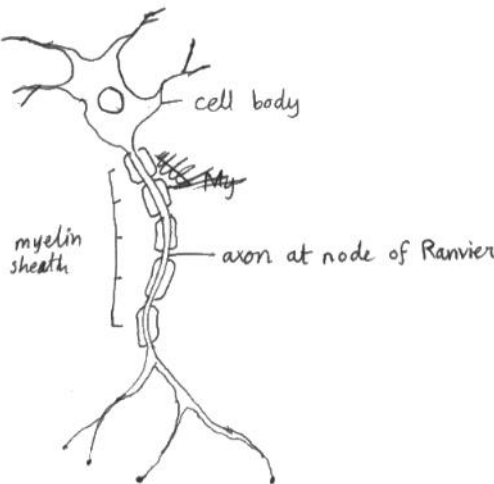


| Question Number | Answer                                                         | Additional guidance      | Mark |
|-----------------|----------------------------------------------------------------|--------------------------|------|
| 5(a)            | <ul style="list-style-type: none"> <li>9500 (%) (1)</li> </ul> | Accept $9.5 \times 10^3$ | (1)  |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Additional guidance                                                                                                                                                                                                                                            | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 5(b)(i)         | <p>An explanation that includes four of the following points:</p> <ul style="list-style-type: none"> <li>ALS developed due to high {concentrations / doses / levels} of BMAA (from eating flying foxes) (1)</li> <li>population of flying fox decreased (1)</li> <li>decrease in number of people with ALS (could be) due to {eating fewer flying foxes / eating a different food source / men dying / fewer men} (1)</li> <li>(delay in decrease in people with ALS could be due) time taken for ALS to develop (1)</li> <li>(initial) increase in number of people with ALS could be due to increased {awareness / education / testing / medical diagnosis techniques} (1)</li> </ul> | <p>ACCEPT {BMAA / (neuro)toxin} accumulated in the people<br/>ACCEPT converse<br/>ACCEPT ALS developed due to eating flying foxes with high BMAA</p> <p>ACCEPT the flying fox no longer ate cycad seeds<br/>ACCEPT the population of cycad trees decreased</p> | (4)  |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Additional guidance                                                                                                                                                                                                                                                                                                                      | Mark |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 5(b)(ii)        | <p>An answer that includes three of the following points:</p> <p>Similarity</p> <ul style="list-style-type: none"> <li>• both can produce an image (of the brain and nervous tissue) (1)</li> <li>• both are non-invasive procedures (1)</li> </ul> <p>Difference (max 2)</p> <ul style="list-style-type: none"> <li>• CT uses {x-rays / electron beams} whereas PET {uses {tracers / isotopes} / detects production of gamma rays} (1)</li> <li>• CT is {quicker / cheaper} (than PET scan) (1)</li> <li>• PET scan can show {(metabolic) activity / biochemical changes} (whereas CT scan does not) (1)</li> </ul> | <p>2 max for differences</p> <p>ignore references to 2D / 3D</p> <p>ACCEPT named tracer e.g. (radioactive) glucose</p> <p>ACCEPT PET scan {takes longer / more expensive} (than a CT scan)</p> <p>ACCEPT PET scan can {show which parts of the brain are active / assess brain activity / show brain function} (whereas CT does not)</p> | (3)  |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                                         | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 5(c)(i)         | <p>An answer that includes the following points:</p> <ul style="list-style-type: none"> <li>• myelin sheath with nodes of Ranvier drawn (1)</li> <li>• cell body drawn at one end of the neuron and synaptic terminal at the other end (1)</li> <li>• two correct labels on the diagram (1)<br/>label arrows MUST BE touching – otherwise ignore it and do not mark that label<br/>accept bracket labels for cell body and { myelin sheath / axon}<br/>incorrect label negates mp3</li> </ul> | <p>Example of diagram</p>  <p>accept synaptic knobs / axon terminals for synaptic terminal<br/>accept axon if label goes along length of myelin sheaths<br/>accept soma for cell body<br/>Schwann cells for myelin</p> <p>max 2 if part of neuron missing eg. no synaptic terminals</p> <p>max 2 if {sensory / relay} neuron drawn max 2 mps 1 and 3</p> | (3)  |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                                         | Additional guidance                                                                                                                                                                                                   | Mark |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 5(c)(ii)        | <p>An explanation that includes three of the following points:</p> <ul style="list-style-type: none"> <li>• (voltage-gated) sodium channels {do not open / are closed / are blocked} (1)</li> <li>• (sufficient) sodium ions cannot enter (neurone) (1)</li> <li>• (axon) membrane cannot be (fully) {depolarised / reach +50mV} (1)</li> <li>• no action potential (generated) (1)</li> </ul> | <p>ACCEPT {no / less} influx of Na<sup>+</sup><br/>do not accept sodium unqualified.</p> <p>ACCEPT threshold potential is not reached</p> <p>ignore action potential inhibited<br/>ignore fewer action potentials</p> | (3)  |